WORKED SOLUTION

STEP I 2003, Question 14

Probability and Statistics • Poisson Distribution

STEP I 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q14

The key idea

The number of friends Jane sees is not the number who call, but that number capped at 2. A capped variable only cares about three cases: nobody calls, exactly one calls, or two or more call. So we never need to sum the whole Poisson series. We just need $P(0)$ and $P(1)$, and "two or more" is whatever probability is left over. For the new friend, we condition on whether they call.

Solution

The original friends

Let N be the number of friends who call on a given day, so $N \sim \mathrm{Po}(2)$ and

$$P(N = k) = \frac{\mathrm{e}^{-2}2^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Jane sees $S = \min(N, 2)$ friends. So $S = 0$ when $N = 0$, $S = 1$ when $N = 1$, and $S = 2$ when $N \geq 2$. Now

$$P(N = 0) = \mathrm{e}^{-2}, \qquad P(N = 1) = 2\mathrm{e}^{-2}, \qquad P(N \geq 2) = 1 - 3\mathrm{e}^{-2},$$

and therefore

$$\mathrm{E}(S) = 0 \cdot \mathrm{e}^{-2} + 1 \cdot 2\mathrm{e}^{-2} + 2\left(1 - 3\mathrm{e}^{-2}\right) = 2 - 4\mathrm{e}^{-2},$$

as required.

A neater way to write the same calculation, which we use again below, is this. For any random variable X taking values in $\{0, 1, 2, \ldots\}$,

$$\mathrm{E}(\min(X, 2)) = 2 - 2P(X = 0) - P(X = 1),$$

because $\min(X, 2)$ falls short of 2 by 2 when $X = 0$, by 1 when $X = 1$, and by nothing otherwise. With $X = N$ this gives $2 - 2\mathrm{e}^{-2} - 2\mathrm{e}^{-2}$ again.

With the new friend

Let $B = 1$ if the new friend calls and $B = 0$ otherwise, so $P(B = 1) = p$. The question says B is independent of N. The total number of callers is now $M = N + B$, and Jane sees $\min(M, 2)$ friends.

? **Ask yourself** Do we need the full distribution of M?

No. By the identity above, we only need $P(M = 0)$ and $P(M = 1)$. Using independence to multiply probabilities:

$$P(M = 0) = P(N = 0)\,P(B = 0) = (1 - p)\mathrm{e}^{-2},$$
$$P(M = 1) = P(N = 1)\,P(B = 0) + P(N = 0)\,P(B = 1) = 2(1 - p)\mathrm{e}^{-2} + p\,\mathrm{e}^{-2}.$$

Hence

$$\begin{aligned}\mathrm{E}(\min(M, 2)) &= 2 - 2(1 - p)\mathrm{e}^{-2} - 2(1 - p)\mathrm{e}^{-2} - p\,\mathrm{e}^{-2} \\ &= 2 - (4 - 3p)\mathrm{e}^{-2}.\end{aligned}$$

Alternatively, condition on B. If the new friend doesn't call (probability $1 - p$), the average is $2 - 4\mathrm{e}^{-2}$, as before. If they do call (probability p), Jane sees $\min(N + 1, 2)$, which is 1 if $N = 0$ and 2 otherwise, with mean $\mathrm{e}^{-2} + 2(1 - \mathrm{e}^{-2}) = 2 - \mathrm{e}^{-2}$. So the average is

$$(1 - p)(2 - 4\mathrm{e}^{-2}) + p(2 - \mathrm{e}^{-2}) = 2 - (4 - 3p)\mathrm{e}^{-2}.$$

So Jane now sees, on average, $2 - (4 - 3p)\mathrm{e}^{-2}$ friends a day, an increase of $3p\,\mathrm{e}^{-2}$.

Checks

- When $p = 0$, the new friend never calls and we recover $2 - 4\mathrm{e}^{-2} \approx 1.4587$.
- When $p = 1$, we get $2 - \mathrm{e}^{-2} \approx 1.8647$, which is the mean of $\min(N + 1, 2)$, as it should be.
- The answer is increasing in p and always below 2, which makes sense.
- Summing $\min(k + b, 2)$ against the exact distribution, and running 4×10^5 simulated days with $p = 0.3$, both give about 1.580, in agreement with $2 - 3.1\mathrm{e}^{-2}$.

Common slips

- Answering $\mathrm{E}(N) = 2$, or $2 + p$, by forgetting the cap at two friends.
- Treating $N + B$ as Poisson. A Bernoulli variable is not Poisson, so the sum is not Poisson either.
- Missing one of the two ways to get exactly one caller (one old friend and no new friend, or no old friends and the new friend).

Beyond STEP

The identity $\mathrm{E}(\min(X,2)) = P(X \geq 1) + P(X \geq 2)$ is a special case of the tail-sum formula $\mathrm{E}(X) = \sum_{k \geq 1} P(X \geq k)$ for non-negative integer random variables. It is the discrete version of integrating the survival function. Capped means like this one appear in queueing and inventory models, where they give the expected number of customers served when capacity is limited. Grimmett and Welsh, *Probability: An Introduction*, covers the tail-sum formula and conditional expectation, which are the two tools used here.

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